SYNVEC

Five-stage RISC-V FPGA system-on-chip with a matrix accelerator

Project overview

SYNVEC is a 32-bit RISC-V-based system-on-chip implemented in SystemVerilog on the Digilent Basys 3 (Artix-7 FPGA). The design combines a five-stage pipelined CPU, on-chip memory, UART communication, programmable PWM, board I/O and a 4 × 4 matrix-multiplication accelerator.

The CPU executes firmware stored in block RAM. A PC sends commands over the board's USB-UART connection, and firmware uses memory-mapped registers to operate the peripherals and return results.

System design

Processor	Five-stage in-order pipeline: fetch, decode, execute, memory and writeback; forwarding, branch handling and load-use stalls.
Memory	16 KiB instruction memory and 16 KiB data memory using FPGA block RAM.
Matrix accelerator	Iterative signed 4 × 4 multiplication with 16-bit input elements and 32-bit results.
Communication	115,200 baud UART with firmware-managed binary commands and responses.
Board I/O	LEDs, switches, seven-segment display and configurable PWM output.

Implementation and debugging

The main design challenge was meeting the 100 MHz clock target after moving to the pipelined CPU. The first routed pipeline missed setup timing by 4.537 ns. Changes to the instruction-fetch/redirect path and load-use handling reduced the critical path, and post-route physical optimization completed timing closure.

Another debugging issue came from the simulation checker: it initially compared inactive byte lanes during a byte store. The checker was corrected to compare only enabled lanes and verify the resulting memory contents.

Project files and tools

The repository contains RTL modules, FPGA pin constraints, the boot ROM, the SystemVerilog testbench, firmware utilities, a verified reference bitstream and selected simulation, implementation and board-test results.

Tools used: SystemVerilog, Vivado, XSim, Python, serial terminal utilities and Analog Discovery 3 / WaveForms.

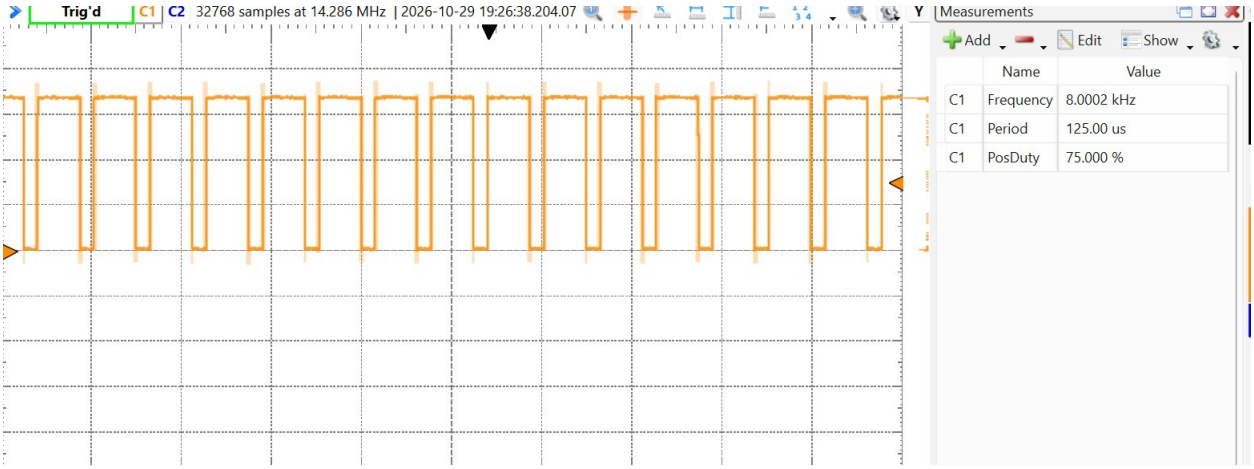
Testing and results

Vivado 2025.2 · XSim · Digilent Basys 3 · Analog Discovery 3

Verification summary

Simulation	The SYNVEC testbench passed two matrix cases, checking 32 result words, UART transfers, LED/PWM commands and memory-mapped accesses.
Fresh FPGA build	The renamed project passed synthesis and implementation at 100 MHz. Reported routed setup slack was +0.031 ns; hold slack was +0.121 ns.
Board test	The freshly built SYNVEC bitstream was programmed onto the Basys 3. A UART command returned the expected acknowledgement, and the requested LED pattern appeared on the board.
Full-system regression	The previously verified hardware build passed 19 UART transactions, 48 matrix results and three oscilloscope PWM measurements.

Measured PWM output



Oscilloscope capture of the 8 kHz, 75% PWM output on the reference FPGA build.